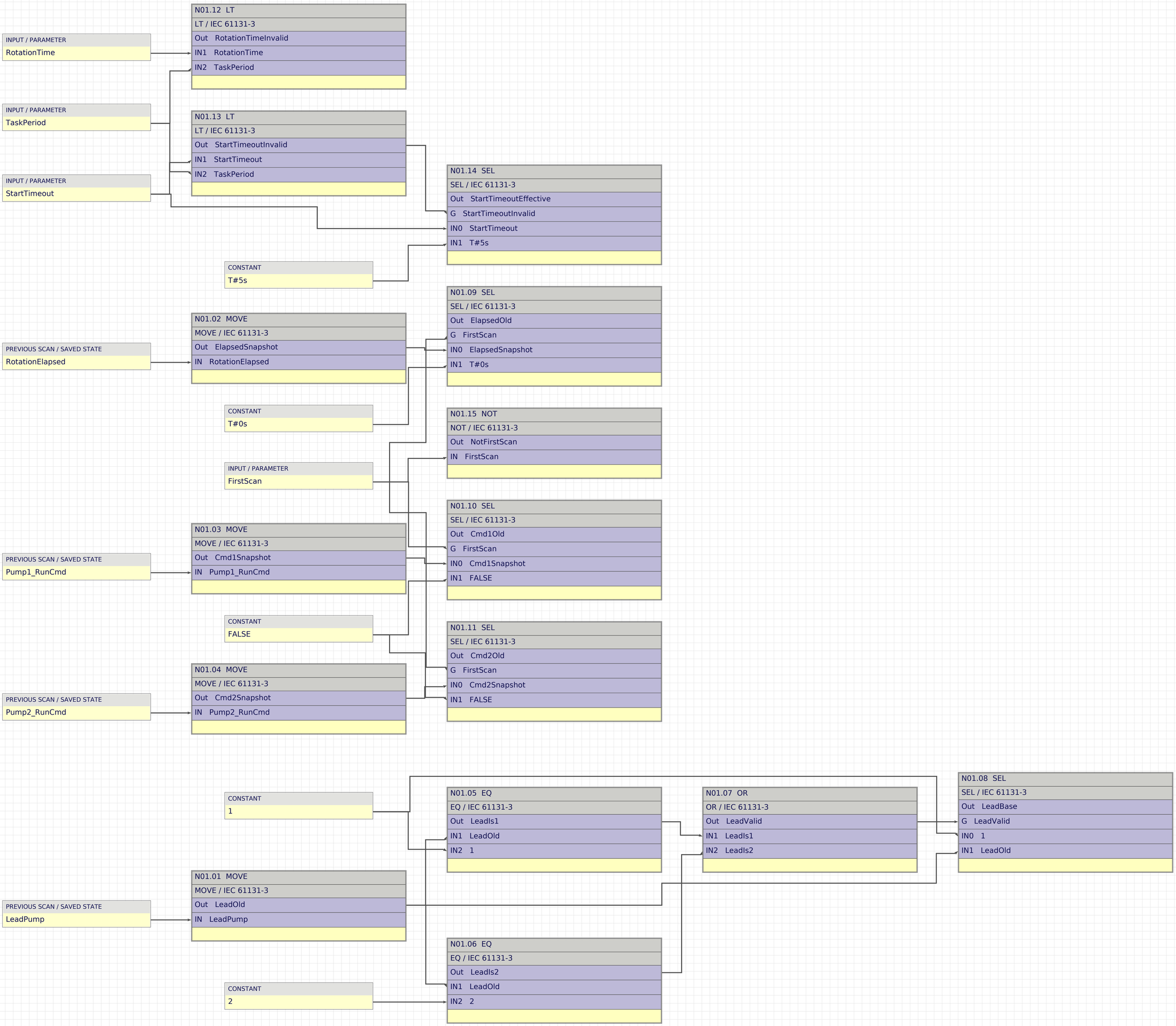
01 | Snapshot / LeadPump normalization / parameter checks

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN | RotationTime: adjustable TIME

TaskPeriod = T#100ms | SEL: FALSE -> IN0, TRUE -> IN1 | numbered blocks define execution order

Static design view: values/status are intentionally not simulated. Logical signals only; no physical I/O mapping.

Execute top to bottom. Snapshot state BEFORE later networks write it. FirstScan is supplied by the PLC platform.



NETWORK 1/9 | IEC logic review diagram in HVAC-style presentation. Not a Schneider project or an online PLC display.

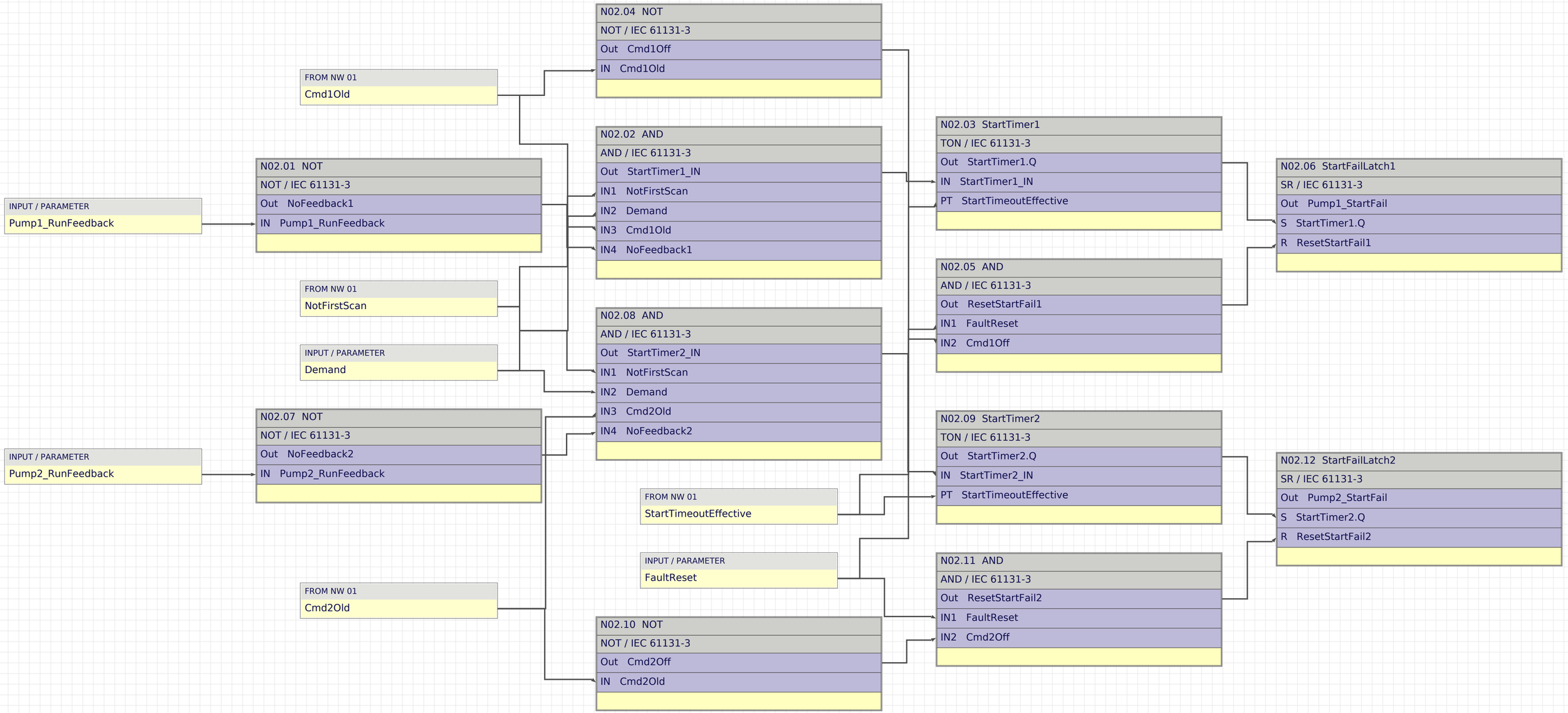
02 | Start / feedback supervision and latched StartFail

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN | RotationTime: adjustable TIME

TaskPeriod = T#100ms | SEL: FALSE -> IN0, TRUE -> IN1 | numbered blocks define execution order

Static design view: values/status are intentionally not simulated. Logical signals only; no physical I/O mapping.

SR is SET-dominant. Reset is accepted only when the previous RUN command is OFF. TON also detects sustained feedback loss.



NETWORK 2/9 | IEC logic review diagram in HVAC-style presentation. Not a Schneider project or an online PLC display.

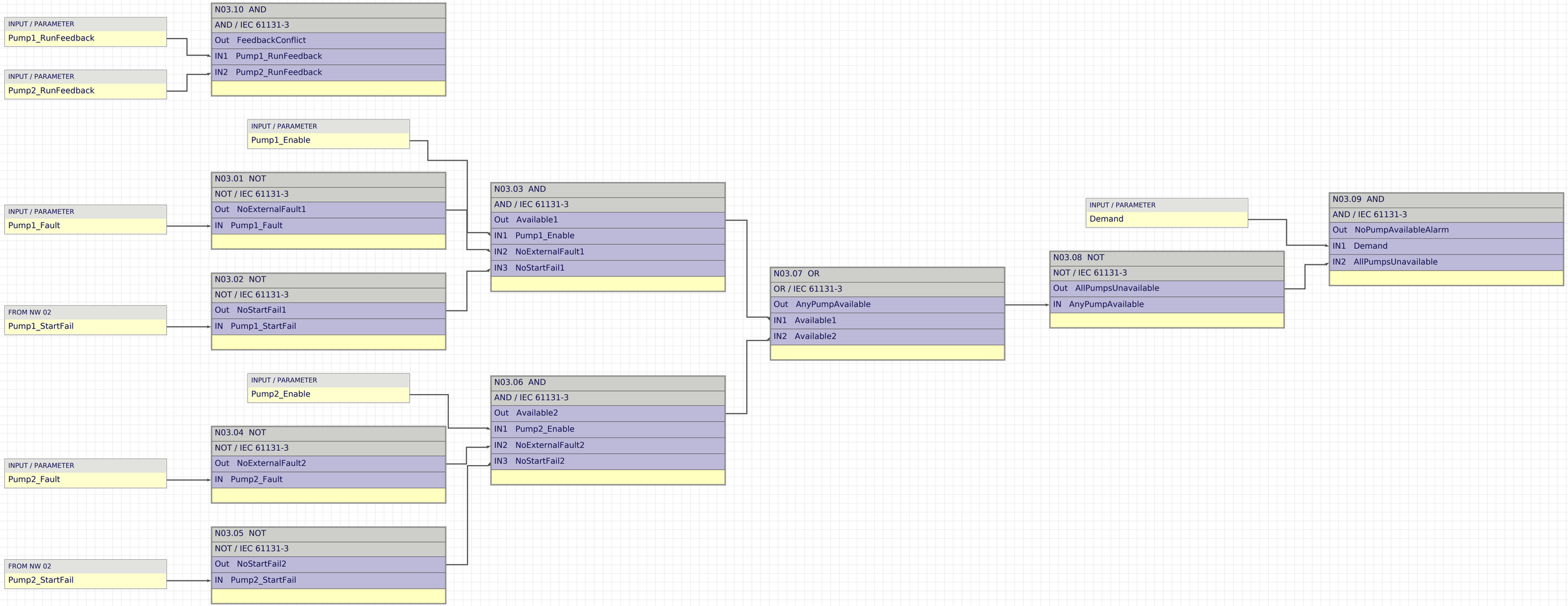
03 | Pump availability / unavailable alarms / feedback conflict

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN | RotationTime: adjustable TIME

TaskPeriod = T#100ms | SEL: FALSE -> IN0, TRUE -> IN1 | numbered blocks define execution order

Static design view: values/status are intentionally not simulated. Logical signals only; no physical I/O mapping.

AllPumpsUnavailable is independent of Demand. NoPumpAvailableAlarm requires Demand.



NETWORK 3/9 | IEC logic review diagram in HVAC-style presentation. Not a Schneider project or an online PLC display.

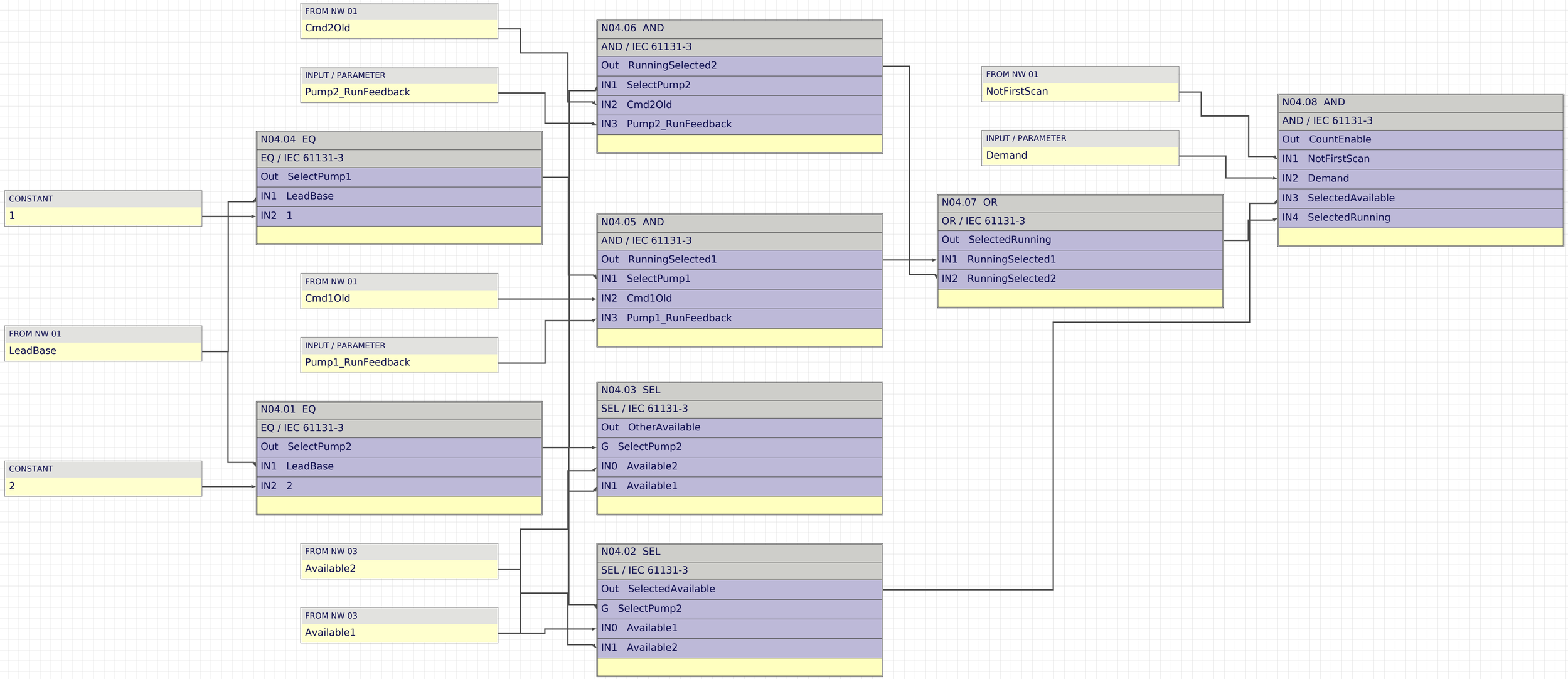
04 | Selected / standby pump and confirmed-running condition

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN | RotationTime: adjustable TIME

TaskPeriod = T#100ms | SEL: FALSE -> IN0, TRUE -> IN1 | numbered blocks define execution order

Static design view: values/status are intentionally not simulated. Logical signals only; no physical I/O mapping.

Time accumulation requires previous RUN command AND the matching RunFeedback. RunCmd alone is insufficient.



NETWORK 4/9 | IEC logic review diagram in HVAC-style presentation. Not a Schneider project or an online PLC display.

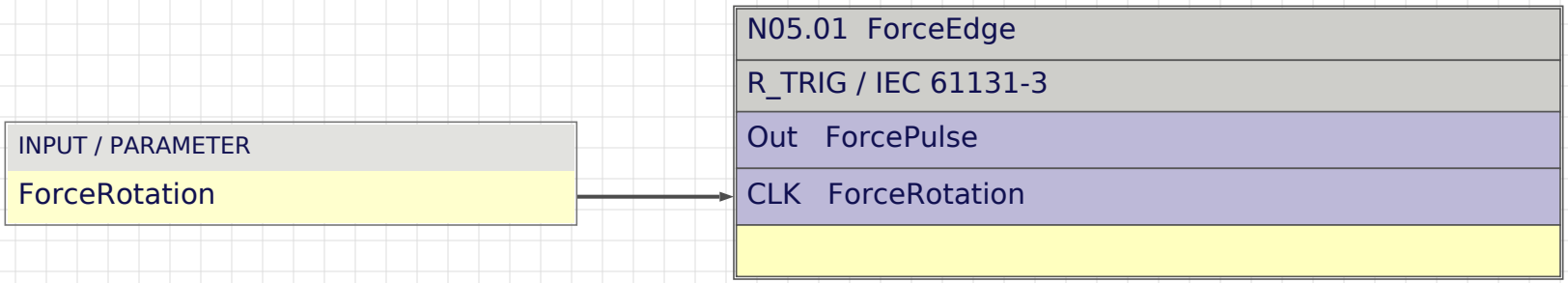
05 | ForceRotation rising edge

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN | RotationTime: adjustable TIME

TaskPeriod = T#100ms | SEL: FALSE -> IN0, TRUE -> IN1 | numbered blocks define execution order

Static design view: values/status are intentionally not simulated. Logical signals only; no physical I/O mapping.

One pulse per rising edge. A request rejected because the standby pump is unavailable is NOT queued.



NETWORK 5/9 | IEC logic review diagram in HVAC-style presentation. Not a Schneider project or an online PLC display.

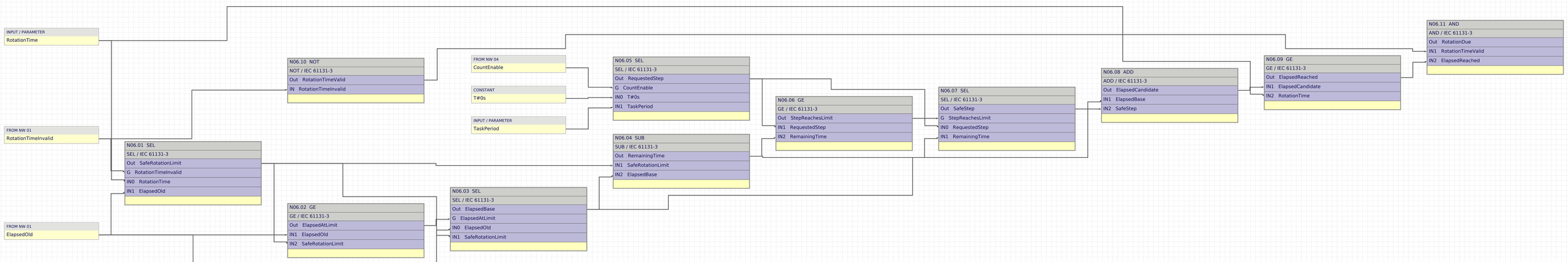
06 | Accumulated rotation timer / TIME arithmetic / RotationDue

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN | RotationTime: adjustable TIME

TaskPeriod = T#100ms | SEL: FALSE -> IN0, TRUE -> IN1 | numbered blocks define execution order

Static design view: values/status are intentionally not simulated. Logical signals only; no physical I/O mapping.

100 ms cyclic task; TaskPeriod = T#100ms. Demand gaps HOLD time. Saturation prevents overflow. Invalid RotationTime disables timed rotation.



NETWORK 6/9 | IEC logic review diagram in HVAC-style presentation. Not a Schneider project or an online PLC display.

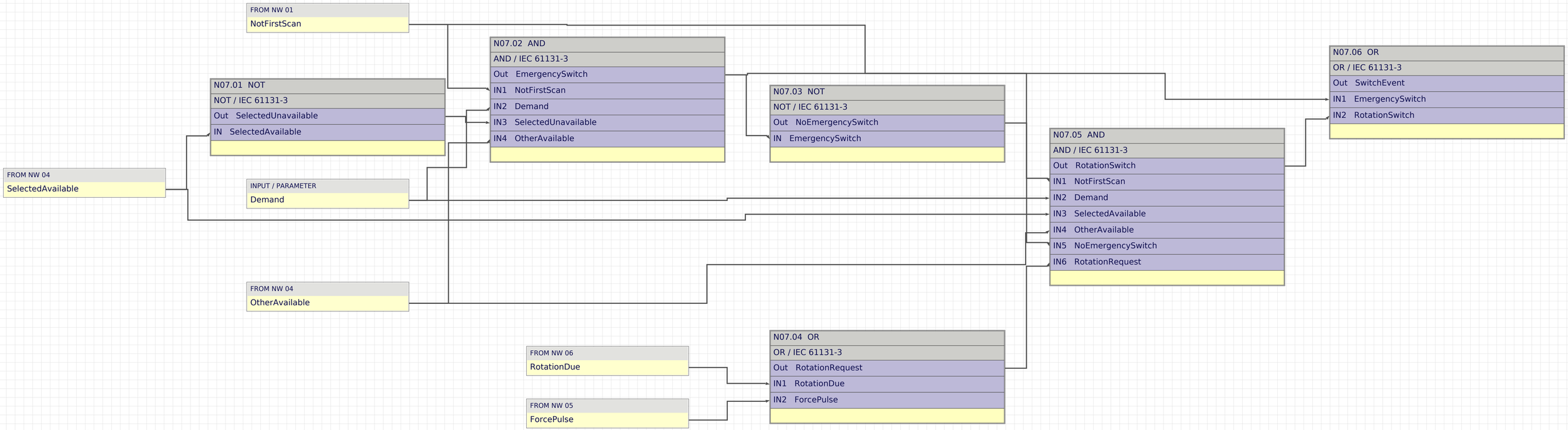
07 | Emergency switching / timed and forced rotation arbitration

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN | RotationTime: adjustable TIME

TaskPeriod = T#100ms | SEL: FALSE -> IN0, TRUE -> IN1 | numbered blocks define execution order

Static design view: values/status are intentionally not simulated. Logical signals only; no physical I/O mapping.

Emergency switching is independent of the rotation timer. Only one switch event is evaluated from LeadBase per scan.



NETWORK 7/9 | IEC logic review diagram in HVAC-style presentation. Not a Schneider project or an online PLC display.

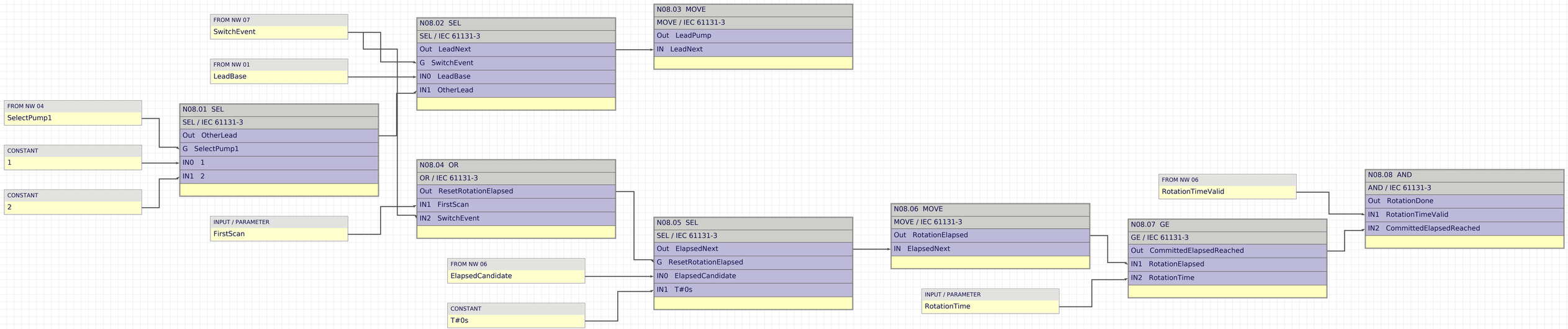
08 | Commit LeadPump / RotationElapsed reset or hold / RotationDone

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN | RotationTime: adjustable TIME

TaskPeriod = T#100ms | SEL: FALSE -> IN0, TRUE -> IN1 | numbered blocks define execution order

Static design view: values/status are intentionally not simulated. Logical signals only; no physical I/O mapping.

LeadPump is RETAIN. RotationElapsed is NOT RETAIN: reset on FirstScan or switching; otherwise commit the accumulated value.



NETWORK 8/9 | IEC logic review diagram in HVAC-style presentation. Not a Schneider project or an online PLC display.

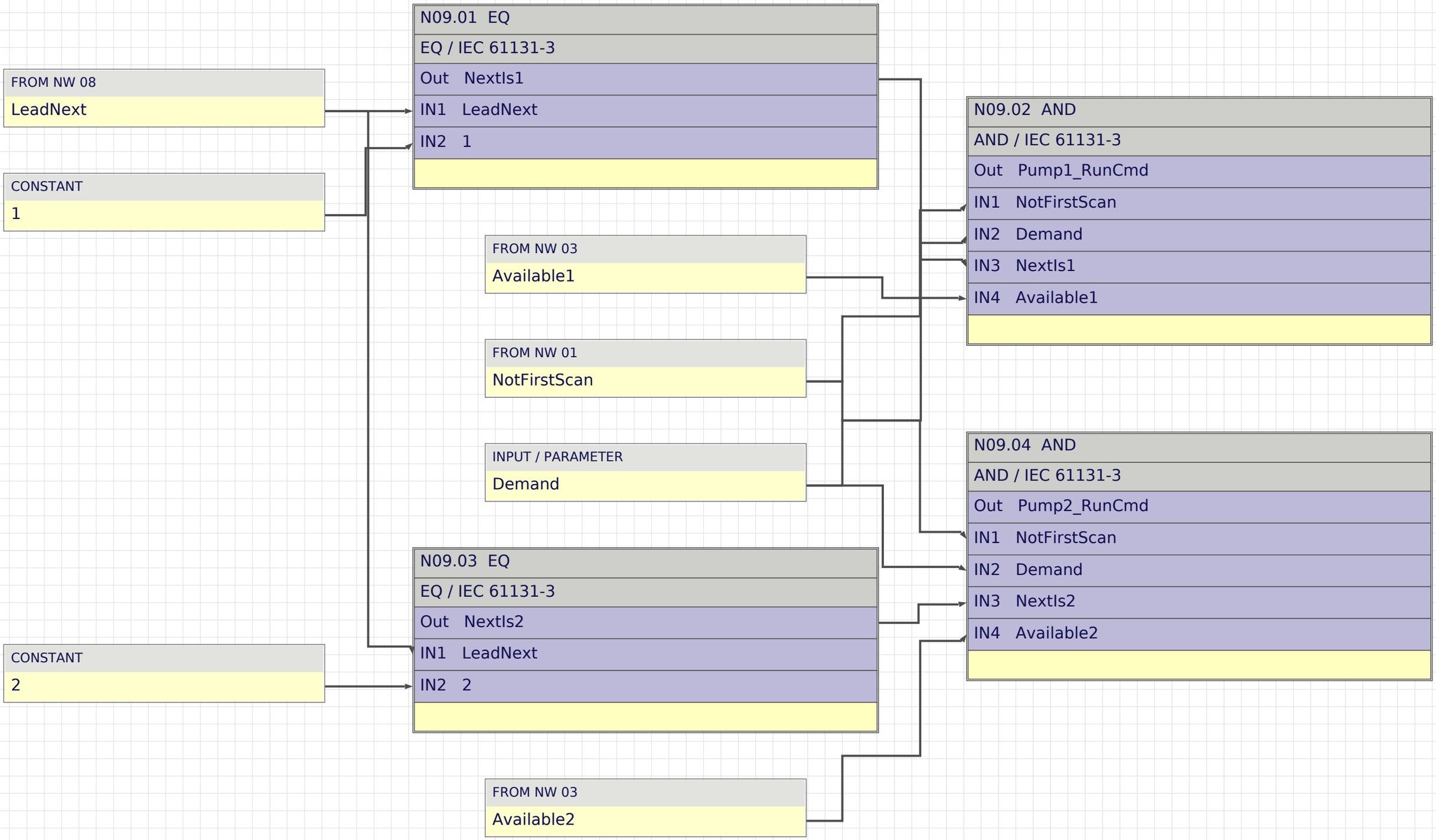
09 | RUN outputs - exclusive single assignment

LeadPump = 1 -> Pump1 | LeadPump = 2 -> Pump2 | LeadPump RETAIN | RotationTime: adjustable TIME

TaskPeriod = T#100ms | SEL: FALSE -> IN0, TRUE -> IN1 | numbered blocks define execution order

Static design view: values/status are intentionally not simulated. Logical signals only; no physical I/O mapping.

The ONLY writers of Pump1_RunCmd and Pump2_RunCmd. EQ(LeadNext,1) and EQ(LeadNext,2) cannot both be TRUE.



NETWORK 9/9 | IEC logic review diagram in HVAC-style presentation. Not a Schneider project or an online PLC display.